MODULE IV · WEEK 12 · LLMS IN THE RESEARCH PIPELINE

Synthetic agents: silicon samples, simulation, ethics

What the literature has established — and a calibration-first study from Turkey

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TODAY

Two halves

I	**The claim**	Persona conditioning, algorithmic fidelity, and what the first wave of studies showed.
II	**The critique**	Variance compression, representational skew, contamination, drift.
III	**How to evaluate**	Three levels of fidelity — distributional, relational, individual.
IV	**A worked case**	Calibrating on TGSS 2024, then running a synthetic experiment inside it.

The question of the week is not "can a model imitate a person?" but "what can we responsibly read off a simulated population?"

THE IDEA

Conditioning a model on a person

Give the model a description of a respondent; ask it the survey question; record the answer.

Why anyone tried this

- Surveys are expensive, slow, and bounded by what can be asked and when
- Some conditions can never be randomised — a war, a peace process, a regime
- A model can be reset and re-run, so the same "respondent" is observed twice

ALGORITHMIC FIDELITY

Argyle et al. (2023): the degree to which a conditioned model reproduces the response patterns of the subpopulation it was conditioned on.

The term is useful because it is a property of a pipeline and a population — not a property of a model.

"Out of One, Many" (2023) is the reference point; everything since has been an argument about how far the idea travels.

FIVE USES

What people actually do with synthetic agents

Ordered from the most ambitious claim to the most defensible one.

Replacement	Stand in for a survey respondent. The strongest claim, and the weakest evidence.
Augmentation	Fill gaps, boost thin subgroups. Depends entirely on where the gaps are.
Piloting	Test items and vignettes before spending money on human fieldwork.
Simulation	Interactive agents producing macro patterns — agent-based modelling with language.
Extension	Take an existing survey into a condition it could not observe. Modest, and workable.

The study in the second half of this session sits on the last line — and the choice of line is the whole argument.

LITERATURE I

What has been shown to work

Argyle et al. (2023)	Demographic backstories reproduce subpopulation patterns in ANES data
Horton (2023)	Classic behavioural-economics experiments replicate on "homo silicus"
Aher et al. (2023)	Turing experiments recover several classic psychology findings
Park et al. (2023)	Generative agents with memory, reflection and planning
Park et al. (2024)	Agents built from interview transcripts for 1,000 people
Ashokkumar et al. (2026)	469 effects across 70 preregistered experiments; r = 0.85 with observed effects

Note the trajectory: retrodiction, then prediction, then experimental replication. Each step raises the stakes of the validation question.

THE CENTRAL FINDING

Direction travels further than magnitude

Predicted effects correlate strongly with observed ones (r = 0.85) — but run roughly twice as large, and the gap changes across archives.

What this licenses

Claims about sign and about the ordering of effects across groups or conditions.

What it does not

Claims about the size of an effect. Outside a benchmarked setting the scale factor is unknown.

Rafikova & Voronin (2026)

Same pattern for attitudes: gradients hold, absolute levels shift toward liberal-prosocial answers.

The practical rule

Report direction and ordering as findings; report magnitude with an explicit caveat, or not at all.

LITERATURE II

Where it breaks

None of these are edge cases. Expect all of them in any application.

Variance compression

Means are reproduced while within-group heterogeneity flattens. Bisbee et al. (2024); Park et al. (2024).

Caricature

Persona conditioning can amplify stereotypes rather than model a person. Cheng et al. (2023).

Contamination

If the replicated study was in the training data, success is recall, not behavioural validity.

Representational skew

Models sit closer to some demographic groups than others; minority positions are misrepresented. Santurkar et al. (2023).

Response biases

Option ordering, labelling, and distorted entropy in multiple-choice answers. Dominguez-Olmedo et al. (2024).

Analytic flexibility

Persona, sampling, temperature, backbone, framing — reported performance is partly a pipeline artefact. Cummins (2025).

THE AWKWARD PART

A paradox worth sitting with

The information that makes a synthetic respondent accurate is usually information that can only be obtained from the survey it is supposed to replace.

What it rules out

Replacement. If richer conditioning is what buys fidelity, and the conditioning comes from a survey, then no survey has been replaced.

What it opens up

Extension. An existing survey can be carried into conditions it never observed — which is exactly what the case study does.

HOW TO EVALUATE

Three levels of fidelity

Most of the confusion in this literature comes from mixing these up.

1

Distributional

Do the marginals match? Means, shares, and — crucially — variance. Wasserstein, Jensen–Shannon, KS.

2

Relational

Are relationships preserved? Correlations, regression coefficients, subgroup gaps in sign and size.

3

Individual

Is this person's answer right? Cohen's κ, weighted κ, per-case recall. The hardest level; most studies do not claim it.

A protocol can win at level 1 and lose at level 3. Which level you need depends on the claim you intend to make.

PART II · A WORKED CASE

A calibration-first synthetic experiment with LLM agents

Simulating public reactions to protest in Turkey

TGSS 2024 · 2,615 respondents · within-persona design

THE DESIGN MOVE

Calibration first, experiment second

THE USUAL SEQUENCE

Build a synthetic population, then test whether it reproduces a known experimental effect.

This tests a joint hypothesis. A failure cannot distinguish a limit of synthetic agents from a badly specified persona, prompt, or model — and a success does not transfer to another population.

THIS STUDY

Make the pipeline itself the research task. Before estimating anything, ask which inference protocol produces the synthetic population closest to the observed one.

Calibration is a precondition for reading a synthetic experiment, not a robustness check appended to it.

DATA

TGSS 2024 as both material and benchmark

Nationally representative face-to-face survey, 2,615 adults, 12 NUTS-1 regions.

Persona construction

- 117 curated variables in six thematic groups
- Demographics · belief & identity · economic position · social values · social-psychological · political attitudes
- Each variable becomes one short natural-language line; labels fixed once and reused
- 11 configurations (C0–C10) × 2,615 respondents = 28,765 profiles

Held-out calibration outcomes

- pacdemons — attended a demonstration in the past year. Binary, ~5% affirmative, n = 1,707
- womenwork — family life suffers when a woman works full time. Five-point Likert, n = 2,588
- Both excluded from every persona, so no respondent's own answer is visible to the model
- Chosen because they differ in response structure — a rare behaviour and an ordinal attitude

CALIBRATION DESIGN

A sequential funnel, fixed in advance

A full factorial would need thousands of runs; the funnel narrows the candidate set at each stage.

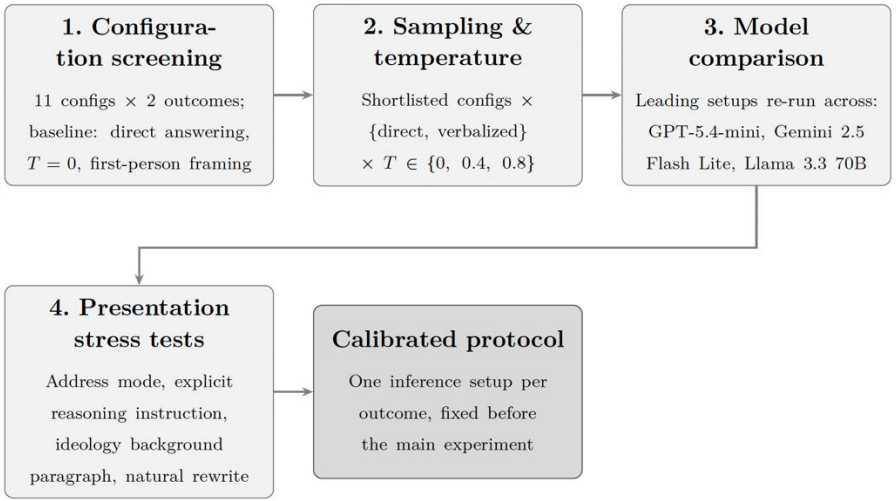


Figure 1: Sequential calibration funnel. Each stage evaluates a different component of the inference setup and narrows the candidate set based on distributional and respondent-level performance. Protocol selection is outcome-specific and reflects the relative informativeness of these metrics at each stage. Stochastic stability across seeds is assessed after protocol selection and reported in Appendix F.

The order is fixed before any results are seen, so the selected protocol follows from a stated procedure rather than from inspecting outcome data.

SCORING

Different response structures, different metrics

BINARY · pacdemons

- Distributional: Jensen–Shannon divergence, prevalence difference Δp
- Respondent level: Matthews correlation, positive-class recall, Cohen's κ
- Why recall matters: a protocol can match a 5% prevalence while recovering almost none of the actual demonstrators

ORDINAL · womenwork

- Distributional: Wasserstein-1 distance, which respects the ordered scale
- Respondent level: quadratic weighted kappa
- Here the aggregate distribution is itself the estimand, so distributional fit carries more weight

No single scalar orders the candidate set across both outcomes. Every cell is scored on both dimensions, and selection is made separately for each.

STAGE 1

Persona content: demographics are not enough

Eleven configurations at a common baseline — direct answering, T = 0, GPT-4o-mini, seed 0.

pacdemons (binary)

C0 (demographics only)	**recall 0.012**
C6	**recall 0.512**
C8	**recall 0.512**
C2	**recall 0.452**

JSD stays below 0.004 in nine of eleven cells, so aggregate fit barely separates them. Recall does.

womenwork (5-point)

C0 (demographics only)	**Wass 0.939**
C10	**Wass 0.493**
C3	**Wass 0.624**
C2	**Wass 0.701**

Every configuration overshoots the observed mean of 3.24. Persona content moves the distribution a long way.

Shortlist carried forward: C1, C2, C3, C5, C6, C10 — the union of the two outcome-specific shortlists.

STAGE 2

Sampling interacts with response structure

Direct answering returns one category; verbalized sampling elicits a distribution and draws from it.

BINARY: DIRECT WINS

Direct answering keeps JSD near zero at every temperature. C6 reaches its highest positive-class recall of 0.536 at T = 0.8.

Verbalized sampling pushes JSD up by an order of magnitude and drags recall down. Spreading probability mass hurts a rare class.

ORDINAL: VERBALIZED WINS

Verbalized sampling improves Wasserstein distance in every configuration — C10 falls from 0.493 to 0.181.

But the gain in distributional fit is not matched by respondent-level agreement: C10 has the best Wasserstein and nearly the worst κw.

This is the fit–agreement trade-off in its clearest form, and it is why several configurations are carried forward rather than one.

STAGE 3

No backbone wins on both criteria

A. pacdemons — C6, direct, T = 0.8 (observed positive rate 4.9%)

MODEL	JSD	Δp	REC(+)	PREC(+)
GPT-4o-mini	**0.00096**	**+1.7**	**0.536**	**0.398**
GPT-5.4-mini	0.00035	+1.0	0.440	0.366
Llama 3.3 70B	0.00167	−1.9	0.310	0.500
Gemini 2.5 Flash Lite	0.02309	−4.9	0.012	1.000

B. womenwork — verbalized sampling, T = 0.8 (observed mean 3.24)

MODEL	CONFIG	WASS	KW	SIM. MEAN
Llama 3.3 70B	C6	0.134	0.212	3.21
GPT-4o-mini	C10	0.181	0.050	3.25
GPT-5.4-mini	**C6**	**0.211**	**0.304**	**3.19**
Gemini 2.5 Flash Lite	C6	0.510	0.230	2.73

STAGE 4

Prompt presentation, and one surprise

Four variants tested against the baseline framing for each selected setup.

VARIANT	pacdemons	womenwork	READING
Second person	recall 0.536 → 0.488	κw 0.304 → 0.299	No gain on either outcome
Reasoning instruction	recall 0.536 → 0.369	Wass 0.211 → 0.155	Aggregate fit up, recall down by a third
Ideology background	recall unchanged	κw 0.304 → 0.317	The only variant that helps without cost
Natural rewrite	recall 0.536 → 0.405	κw 0.304 → 0.257	Fluent biography weakens both criteria

The ideology background paragraph is a short description of the two-dimensional structure of Turkish politics — a left–right political axis and a secular–religious cultural axis. Compact, domain-specific context outperformed generic instruction.

THE RESULT

Two protocols, fixed in advance

COMPONENT	pacdemons (binary)	womenwork (5-point)
Persona configuration	**C6**	**C6**
Sampling strategy	**Direct answering**	**Verbalized sampling**
Temperature	**0.8**	**0.8**
Model	**GPT-4o-mini**	**GPT-5.4-mini**
Address mode	**First person**	**First person**
Prompt framing	**+ ideological background**	**+ ideological background**
Positive-class recall	**0.512**	—
JSD	**0.0011**	—
Wasserstein distance	—	**0.181**
KW	—	**0.312**
n	**1,707**	**2,588**

Same persona, same temperature, same framing. They differ exactly where the response structure differs.

TWO CHECKS

Did the funnel just overfit two items?

TRANSFER TO UNSEEN ITEMS

- Binary protocol on pacvolunteer, paccontact, paccompl: recall 0.498–0.662, JSD 0.0001–0.0059
- Ordinal protocol on famroles, satdem, polint: Wass 0.281–0.399, κw 0.513–0.562
- Neither protocol was adjusted; none of these items entered selection

STOCHASTIC STABILITY

- Binary: 99.1% exact agreement across seed pairs, mean κ = 0.930
- Ordinal: aggregates stable, but only 31.0% exact agreement across seeds (68.2% within one category)
- So the ordinal protocol supports claims about distributions and average contrasts — not about which individual holds which view

Reading synthetic output at the level of aggregation the protocol supports is part of using it honestly.

THE EXPERIMENT

One factor, two levels, the same protest

Political process theory says public reactions depend on the surrounding structure of opportunity and threat — which no experiment can randomise.

THREAT (security)

A defence-ministry statement on intensified armed clashes with Kurdish armed groups in northern Syria, casualties, and heightened security measures.

OPPORTUNITY (peace)

Ongoing talks between the government and Kurdish political representatives, widely attended Newroz celebrations, and supportive messages from party leaders.

Both vignettes end with the same focal event: a march in Ankara demanding Kurdish-language education.

Outcomes · Movement legitimacy (five-point Likert, ordinal protocol) · Behavioral intent — would you personally attend? (binary protocol)

WHY THIS DESIGN IS DIFFERENT

The individual counterfactual

An agent can be returned to its initial state and run again. All 2,615 agents answer both conditions, in independent prompt turns with no shared state.

In a human survey

Each respondent sees one condition. What they would have answered under the other is missing by construction (Holland 1986).

Here

No sample split. Every respondent characteristic is held constant across the comparison, because it is the same persona.

So we settle for

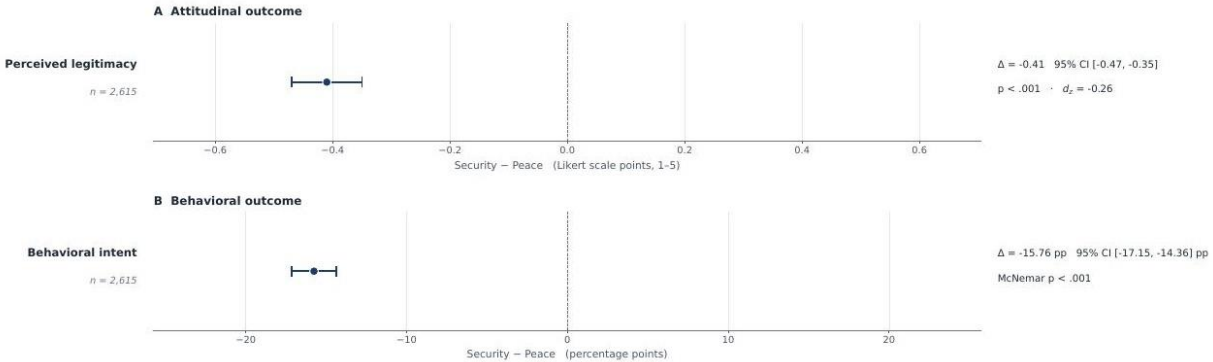
Average effects across people, not effects within them.

The honest caveat

This is the individual counterfactual of the synthetic respondent. Whether it corresponds to the human one is a separate question.

RESULT 1

Political context shifts both outcomes



Perceived legitimacy

−0.41 Likert points under threat
95% CI [−0.47, −0.35] · $p < .001$ · $d_z = -0.26$
Mean 2.47 under opportunity, 2.06 under threat

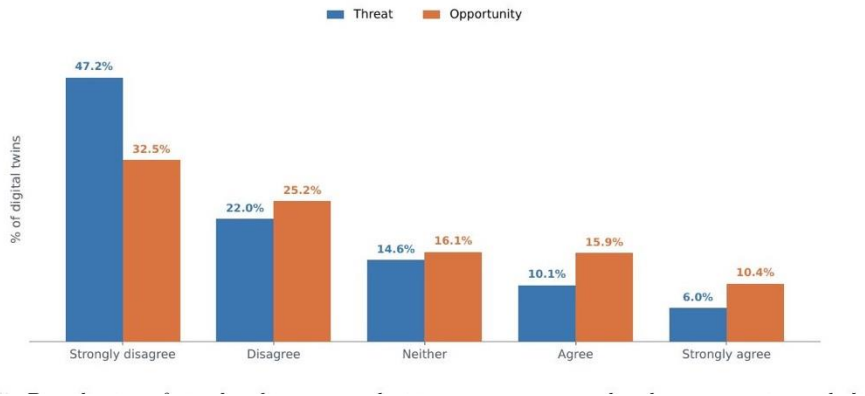
Behavioral intent

−15.76 percentage points under threat
95% CI [−17.15, −14.36] · McNemar $p < .001$
22.5% would attend under opportunity, 6.8% under threat

RESULT 1 (CONT.)

The aggregate shift is a within-persona shift

Full response distributions under the two conditions.



The paired view

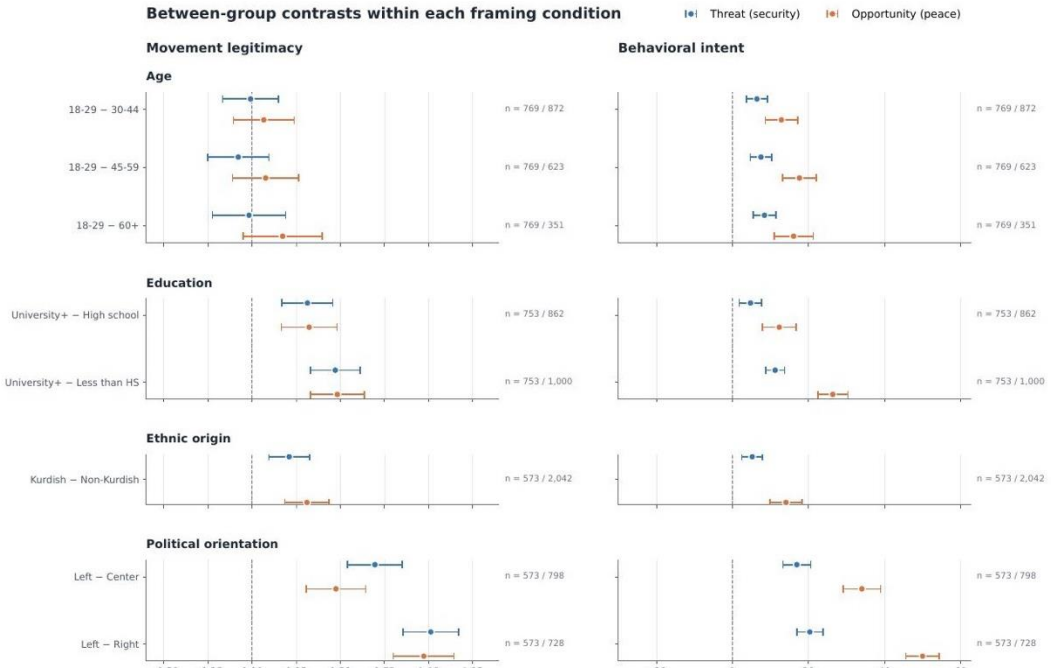
- Legitimacy: 43.7% rate higher under opportunity, 31.5% identical, 24.7% higher under threat
- Intent: 15.8% would attend only under opportunity — and none only under threat
- Lowest legitimacy category rises from 32.5% to 47.2%; the two agreement categories fall from 26.3% to 16.1%

The direction is not carried by a small subgroup; it is a broad shift across the simulated population.

RESULT 2

Between-group contrasts within each condition

Does the synthetic population differentiate — or does it flatten?

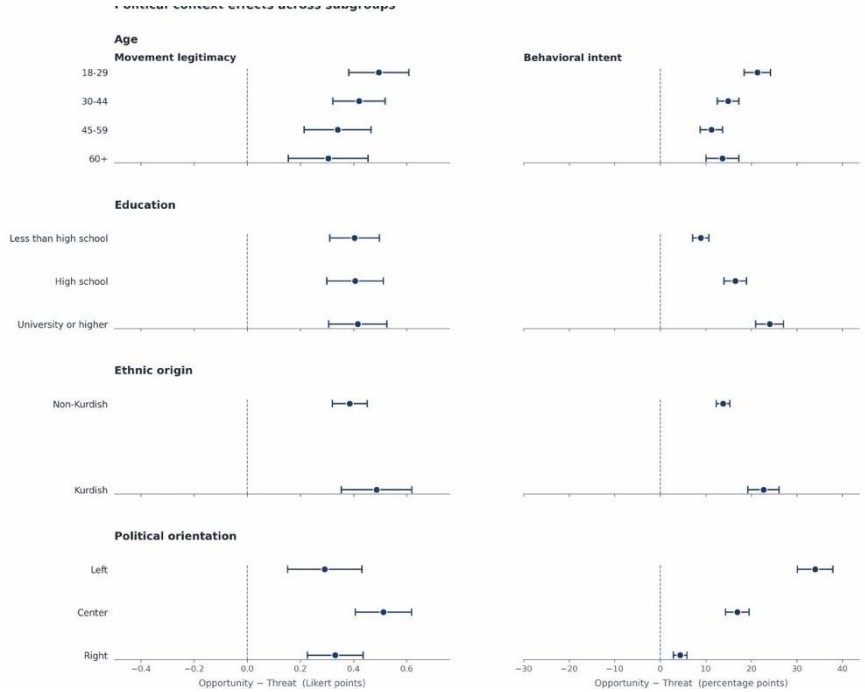


It differentiates

- Left-leaning agents report higher legitimacy and intent than centrists and the right, in both conditions
- Kurdish agents report higher legitimacy and intent than non-Kurdish agents, in both conditions
- University-educated agents show higher behavioral intent
- Age differences are small and unsystematic, especially for legitimacy

RESULT 3

The context effect itself varies by group



Behavioral intent moves most

Opportunity-minus-threat is 34.0 percentage points among left-leaning agents, 16.9 among centrists, and 4.4 among right-leaning agents. The contrast is also larger among Kurdish than non-Kurdish agents, among university-educated than lower-educated, and among younger than older agents.

Legitimacy moves less

The contrast is broadly similar across education levels and somewhat larger among Kurdish agents. Political orientation follows a different pattern here: the largest legitimacy contrast falls among centrists rather than among the left.

INTERPRETATION

Why intent moves more than legitimacy

Sympathy converts into participation only through a sequence of steps at which most sympathisers drop out — and the later steps are governed by the costs of acting, not by evaluation of the cause (Klandermans & Oegema 1987).

The mechanism

A threatening context raises the cost of attending without changing the normative judgment that a language-rights claim is legitimate. Attitudes are stickier; intent absorbs the shift.

Consistent with

Experimentally induced fear reduces expressed willingness to dissent in an authoritarian setting (Young 2019).

And with Turkey-specific work

Conflict casualties shift voters toward hardline positions (Kıbrıs 2011); participation in Kurdish contentious politics is structured by ideological and communal proximity (Tezcür 2016).

The centrist pattern

The largest legitimacy contrast among centrists fits accounts in which the poles are anchored and the middle moves with the political climate (Somer 2019).

WHAT THESE NUMBERS ARE

Epistemic status, stated plainly

WE REGARD AS CREDIBLE

- The direction of the contrast
- The ordering across subgroups
- The asymmetry between the attitudinal and behavioral outcomes

The scale is not.

LIMITS THAT FOLLOW

- Calibration is specific to one survey, one country, one moment
- Conditioning comes from the survey itself, so nothing is replaced
- Selected protocols are the best in a searched region, not global optima
- We cannot verify whether Kurdish agents reflect real reasoning or a pretraining association
- Commercial models change, so reproducibility depends on archived completions

The single most valuable next step is a fielded human benchmark on the same vignettes, which would turn the unknown scale factor into an estimate.

FOR YOUR OWN WORK

A reporting checklist for synthetic populations

1 **Model, version, access date**

Open or closed weights; archive the completions.

2 **The full prompt**

Every variant tried, not only the one that survived.

3 **Temperature, seed, repeats**

And the sampling strategy, if you used one.

4 **Persona variable set**

What went in, and what was deliberately held out.

5 **Calibration / test separation**

Which items scored the protocol, and which never entered it.

6 **Fidelity at all three levels**

Distributional, relational, individual — including a variance report.

7 **Seed stability**

Aggregate and respondent-level agreement across seeds.

8 **What the result does not support**

Say it in the paper, before a referee says it for you.